

Lucas showed how asset prices responded to an economy's factors and that these prices were set by agents' optimal consumption and market clearing. In Mehra and Prescott's adaptation of Lucas, the factor was consumption, and there was a representative agent with constant relative risk aversion (CRRA) preferences (see chapter 2).² Just as the capital asset pricing model prices all assets through exposure through the market factor, consumption-based asset pricing explains the returns of all assets through exposure to the consumption factor. The relation between the consumption factor, though, and how it enters the risk premiums of individual assets is a transformation that depends on the representative agent's degree of risk aversion.

To gain intuition why the returns of equity are anomalously high relative to this simple setting, Table 8.3 reports log returns of the S&P 500 with real consumption growth over June 1947 to December 2010. The Sharpe ratios are approximately 0.3 using either a nominal short rate of 5% or a real short rate of 2%. Table 8.3 shows that the volatilities of equity returns and consumption growth do not line up: equity volatility is high, around 17%, while consumption volatility is low, around 2%. Equity returns are also poorly correlated with real consumption growth, with a correlation of below 15%. Thus, when bad times of low consumption growth occur, equity returns do not tend to fall. Since equities do not exhibit losses when agents experience bad times, as measured by times of slowing consumption, the equilibrium risk premium should be low.

A skeptical reader might now be asking, "So what?" Consumption does not move around very much, and there are better proxies for economic growth (see chapter 7). I have also told you that most agents do not have mean-variance type utility functions, of which CRRA is an example (see chapter 2). Is this equity

Table 8.3

<i>Log S&P 500 Returns, Consumption Growth</i>			
	Nominal	Real Stock	Real Consumption
	Stock Returns	Returns	Growth
Mean	10.4%	7.1%	3.3%
Stdev	16.5%	16.6%	1.7%
Sharpe Ratio	0.328	0.308	
	with 5% nom	with 2% real	

² In the context of factor theory (see chapter 6), Mehra–Prescott's world has a nonlinear pricing kernel, which is given by how the representative agent responds (at the margin) to changes in real consumption growth.